

Problem Solving in Geometry (V)

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1. Let ABC be a triangle, the incircle of triangle ABC touches the side BC at D and let J_B and J_C be the incentres of the triangles ABD and ACD , respectively. Prove that the circumcentre of the triangle AJ_BJ_C lies on angle bisector of $\angle BAC$. (RMM15-4)

2. Let $ABCD$ be a cyclic quadrilateral and let P be a point on the side AB . The diagonal AC intersects the line segment DP at Q . Let K and L be points on the line BC such that $KP \parallel CD$ and $LQ \parallel BD$. Prove that the circumcircles of the triangles BKP and CLQ are tangent. (RMM19-1)

3. Let $ABCD$ be an isosceles trapezium with $AB \parallel DC$. Let E be the midpoint of AC . Denote ω and Ω the circumcircles of the triangles ABE and CDE , respectively. Let P be the point of intersections of the tangent to ω at A and the tangent to Ω at D . Prove that PE is tangent to Ω . (RMM20-2)

4. Let ABC be a triangle and let D be a point on the segment BC , $D \neq B$ and $D \neq C$. The circle ABD meets the segment AC again at an interior point E . The circle ACD meets the segment AB again at an interior point F . Let A' be the reflection of A in the line BC . The lines $A'C$ and DE meet at P , and the lines $A'B$ and DF meet at Q . Prove that the lines AD , BP and CQ are concurrent (or all parallel). (RMM16-1)

5. Given an acute triangle ABC , let H and O be its orthocentre and circumcentre, respectively. Let K be the midpoint of the line segment AH . Also let l be a line through O , and let P and Q be the feet of perpendicular from B and C to l , respectively. Prove that $KP + KQ \geq BC$. (RMM23-4)